



**Unit 2**  
**Science stage 6**  
**Topic 2.1**  
**Properties of**  
**substances**

# Lesson Objectives

By the end of the lesson, I will know

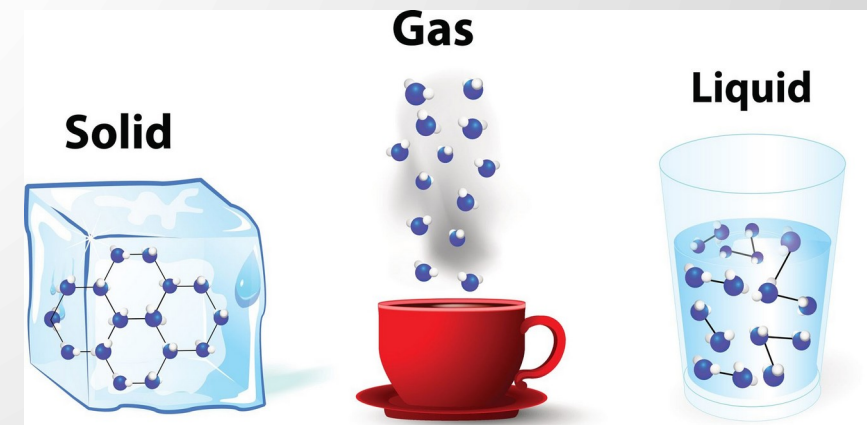
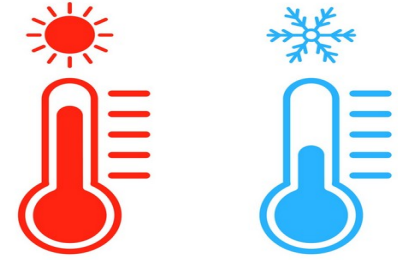
6Cp.01 Know that the temperature at which a substance changes state is a property of the substance.

6Cc.03 Describe the difference between boiling and evaporation.

6Cp.02 Know that gases have properties, including mass.

# Success Criteria

- I can understand that the temperature at which a substance changes state is a property of the substance.
- I can describe the difference between boiling and evaporation.
- I can identify properties of gases.



# New Vocabulary-Notebook

1. Melting points
2. Boiling point
3. Properties
4. Evaporation
5. Mass

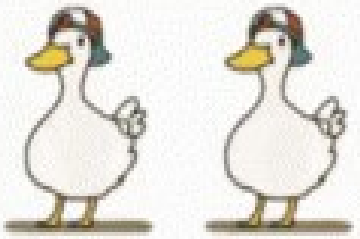
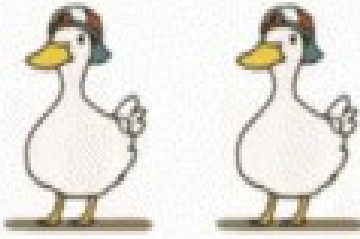
*Please note- add definitions to the new terms using words you can understand*

# 01 Warm up

What is a Gas?

What is a Liquid?

What is a solid?

| SOLID MOLECULES                                                                      | LIQUID MOLECULES                                                                     | GAS MOLECULES                                                                        |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|   |   |   |
|  |  |  |

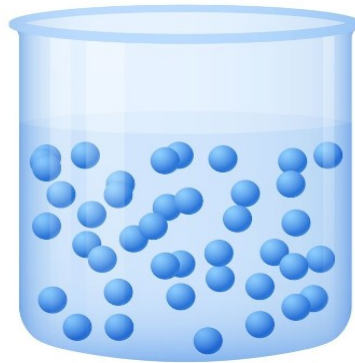
# State of substances

## SOLID



- Rigid
- Fixed Shape
- Fixed Volume
- Cannot be squashed

## LIQUID



- Not Rigid
- No Fixed Shape
- Fixed Volume
- Cannot be squashed

## GAS



- Not Rigid
- No Fixed Shape
- No Fixed Volume
- Can be squashed

# What do you notice?

Explain what is happening to the water?



# Introductory video

**WHAT IS  
FREEZING,  
MELTING AND  
BOILING POINT?**



# Myth Buster!

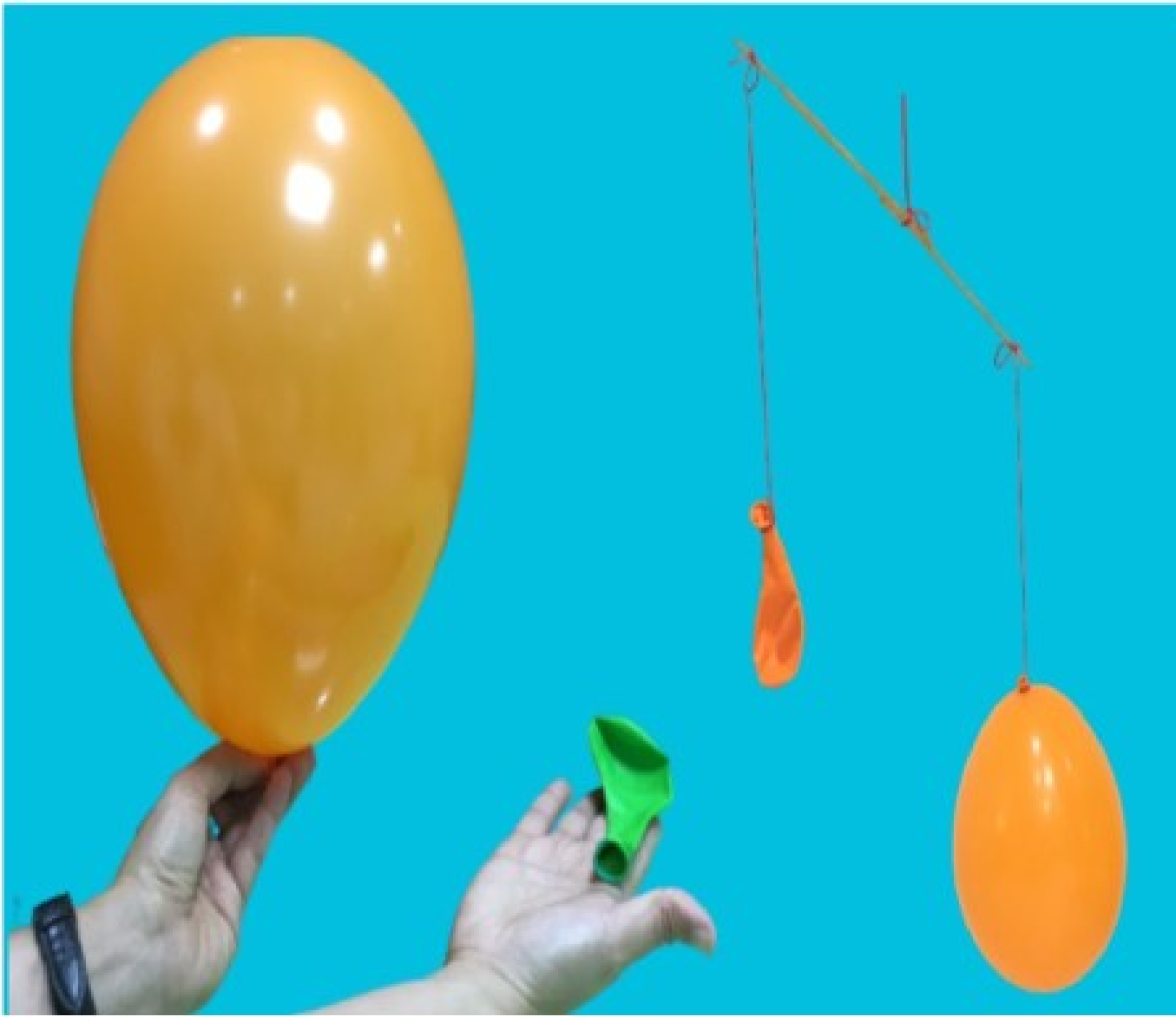


many people make mistakes here!

Do gasses have mass?



to help us to understand it more effectively.



If gas didn't have mass, the blown up balloon would have an equal weight to the empty balloon.

## Mass

We process information to help us to understand it more effectively.

# Melting point

Water has a melting point of  $0^{\circ}\text{C}$  which means

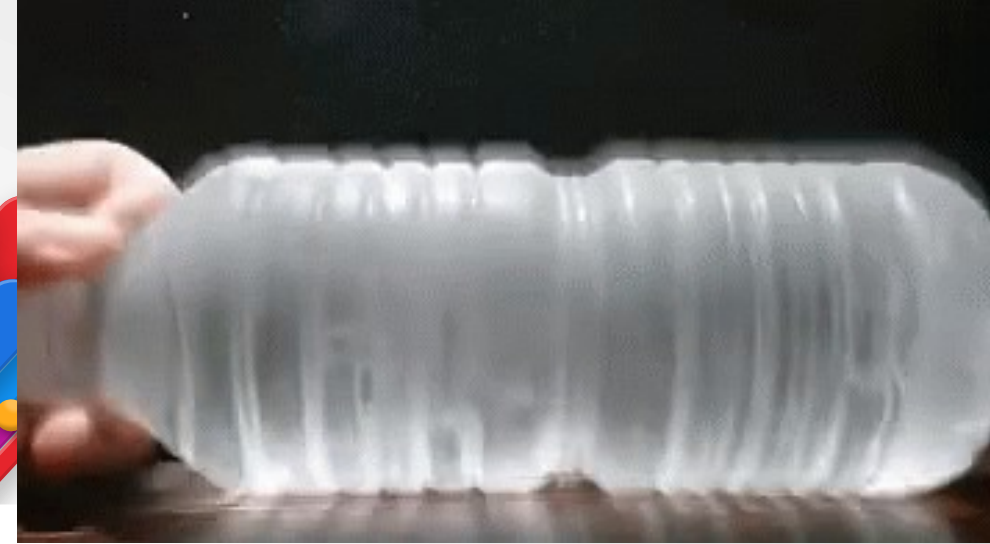
At  $0^{\circ}\text{C}$  water will turn into a liquid when it was frozen

The temperature at which something becomes liquid



# Freezing point

The temperature at which something becomes solid



— Irene M. Pepperberg



# Evaporation

is a substance's change of phase from liquid to gas.



liquid

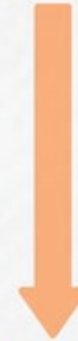


gas

Evaporation is a type of vaporization.



heat

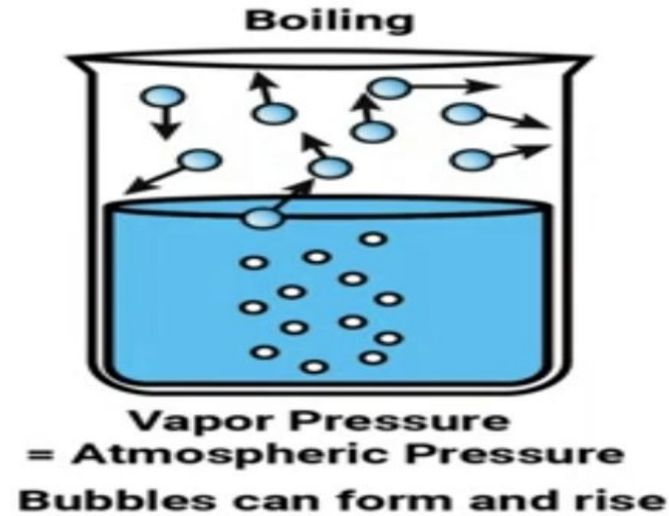
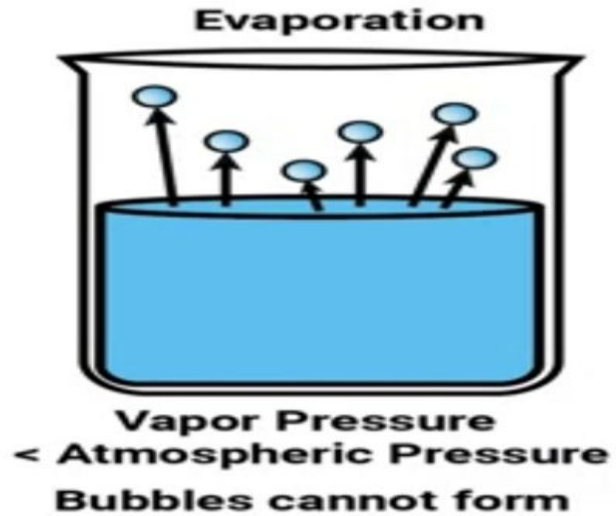


gas

evaporation



# Evaporation Vs Boiling

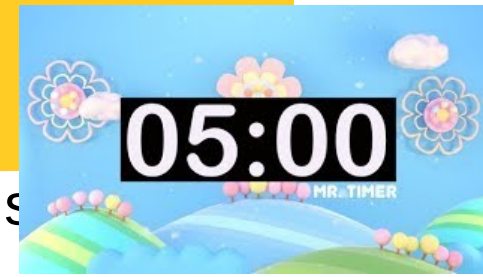


| Evaporation                                   | Boiling                                       |
|-----------------------------------------------|-----------------------------------------------|
| No bubbles. 😊                                 | Bubbles form throughout liquid.               |
| Takes place only on <b>surface</b> of liquid. | Takes place <b>throughout</b> the liquid.     |
| <b>Slower</b> process.                        | <b>Faster</b> process.                        |
| Occurs at <b>any</b> temperature.             | Boils a <b>specific</b> temp.(boiling point). |
| Temperature can <b>fluctuate</b> .            | Temperature remains <b>constant</b> .         |
| Thermal energy from <b>surroundings</b> .     | Thermal energy from <b>external source</b> .  |

## Similarities:

- Both involve liquids changing to gases.
- Both need energy to take place.
- Both are physical changes.

# QUICK TRUE OR FALSE

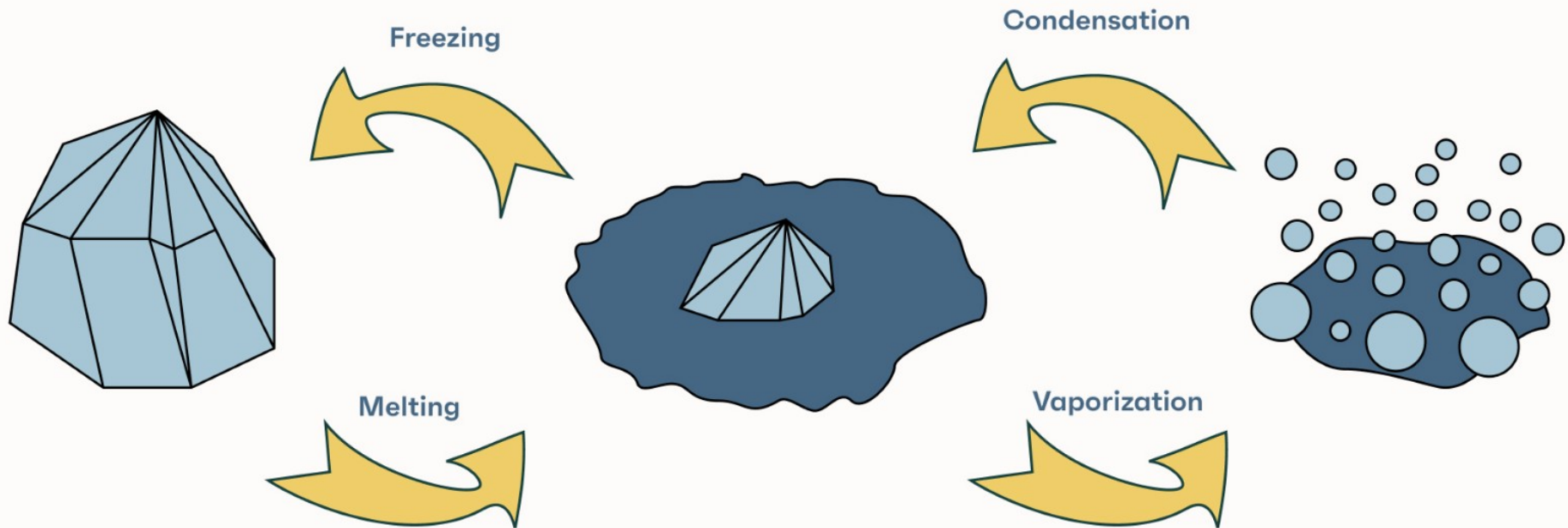


1. The temperature at which a substance changes state is a property of that substance. (False/True)
2. Evaporation only happens when a liquid reaches its boiling point. (False/True)
3. Boiling occurs at a fixed temperature for a given substance. (False/True)
4. Gases do not have mass because we cannot see them. (False/True)
5. Both boiling and evaporation change a liquid into a gas. (False/True)
6. The boiling point of water is always the same under normal conditions. (False/True)
7. Evaporation can happen at any temperature below the boiling point. (False/True)
8. When a balloon is inflated, its mass stays exactly the same as when it was empty. (False/True)

**Solid**

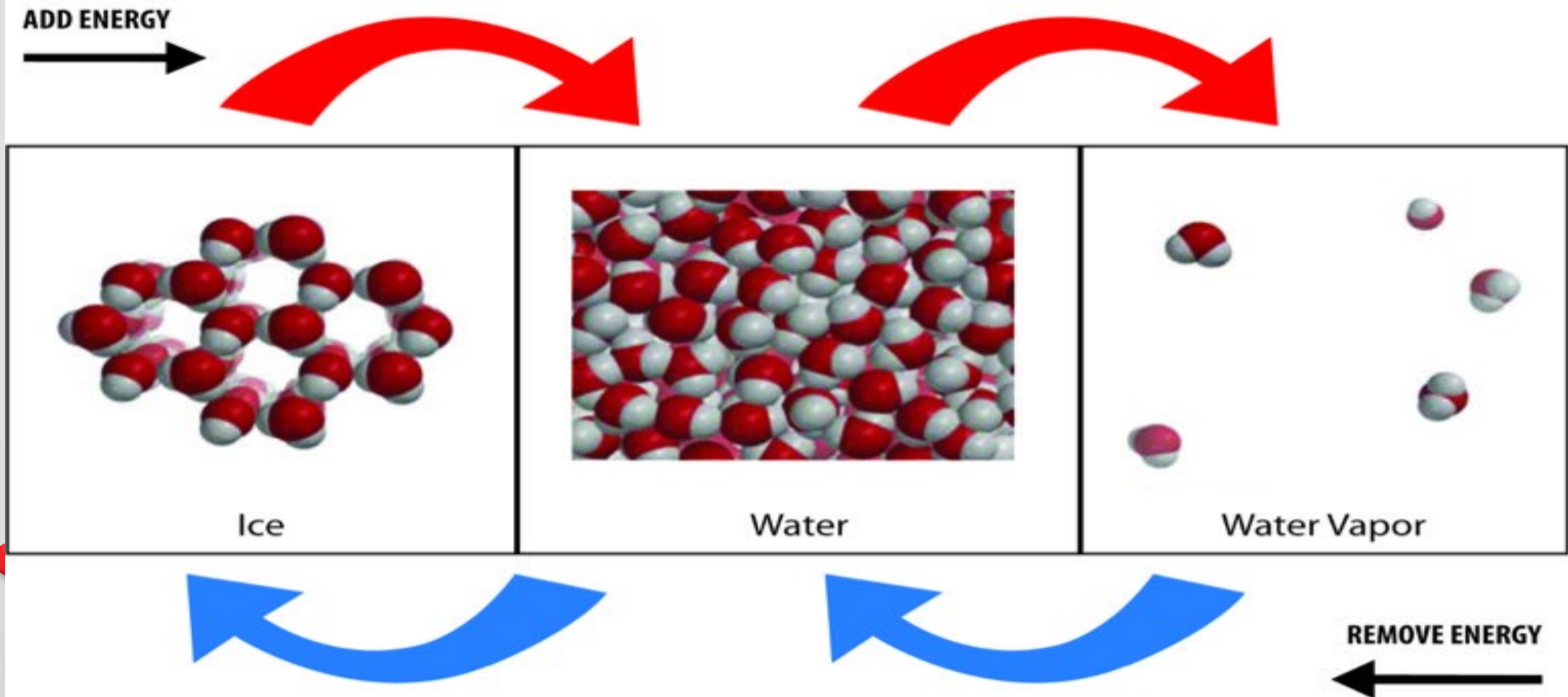
**Liquid**

**Gas**



# Changes of State- Notebook Activity

Write down the name of the changes in the red arrows and the blue arrows



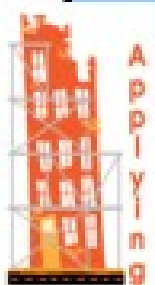


# Data for experiment

## Substances in the experiment

| Substance | Melting Point ( $^{\circ}\text{C}$ ) | Observations (e.g., fast, slow, smoky) |
|-----------|--------------------------------------|----------------------------------------|
| Ice       | $0^{\circ}\text{C}$                  | Melted very quickly into water.        |
| Butter    | $34^{\circ}\text{C}$                 | Melted into a yellow, oily liquid.     |
| Beeswax   | $63^{\circ}\text{C}$                 | Took the longest time to melt.         |

# Comparing Melting Points



## Think like a scientist 1

### Measure and compare melting points

**You will need:** three different solids, three pans, a hotplate, a thermometer, a stopwatch or digital watch

This activity is a teacher demonstration.

**Safety:** Do not touch the hot plate or pan: it can burn you.

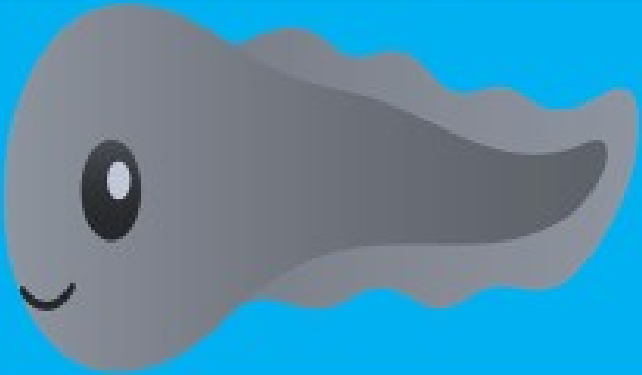
You teacher will do the following:

- 1 Place a solid in a pan.
- 2 Heat the solid substance until it melts. Remove the pan from the hot plate as soon as the substance melts.
- 3 Measure the temperature of the melted substance. You should record the reading in a table.
- 4 Repeat steps 1–3 with the two other solids.



We apply our learning to demonstrate understanding and then reflect on this.

# Differentiation Levels!



## Tadpole Level

- 1 Which substance had:
  - a the highest melting point?
  - b the lowest melting point?
- 2 Draw a bar chart of the results.



## Frog Level

- 3 Write a conclusion for the investigation.  
Remember a conclusion is what you found out in the investigation.
- 4 Would the melting points change if you heated more of the substance in each pan? Explain your answer.
- 5 Why is the melting point of a substance always lower than its boiling point?



# Let's complete a definition match

- Blood
- Conduction
- Pulse
- Properties
- Circulation
- Thermal Conductor
- Boiling Point
- Circulatory System
- Electrical Conductor
- Melting Point

- The temperature at which a liquid becomes a gas.
- The temperature at which a solid becomes a liquid.
- Something about the substance that can be used to describe it.
- The movement of heat or electricity through matter.
- Materials and substances which conduct heat well.
- Materials and substances which conduct electricity well.
- The pumping of blood around the body.
- A small beat that can be felt due to the pressure of blood as the heart pumps blood around the body.
- The system formed by the heart, blood vessels and blood to carry food and oxygen around the body.
- A red liquid that carries food and oxygen to all parts of the body.



**Unit 2**  
**Science stage 6**  
**Topic 2.2**  
**Thermal and**  
**electrical conductors**



# Lesson Objectives

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# New Vocabulary

1. Thermal conductor
2. Electrical conductor
3. Mass

# What are material properties?

**Hard**



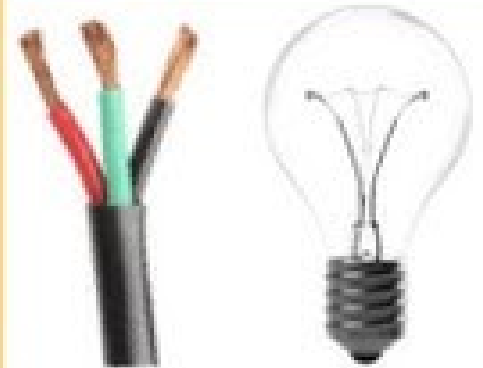
**Soluble**



**Transparent**



**Conductor**



**Soft**



**Insoluble**



**Opaque**



**Insulators**



Pronounce and understand today's keywords

# Properties



Shiny

Hard

Transparent



Soft



Opaque

Dull

We gather all of the information that we need in order to learn.

# Physical Properties

- Can be observed or measured without changing the matter's identity.

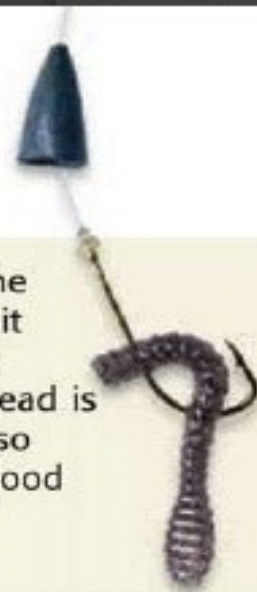
## Examples of Physical Properties



**Thermal conductivity** (KAHN duhk TIV uh tee) is the rate at which a substance transfers heat. Plastic foam is a poor conductor.



**State** is the physical form in which a substance exists, such as a solid, liquid, or gas. Ice is water in the solid state.



**Density** is the mass per unit volume of a substance. Lead is very dense, so it makes a good sinker for a fishing line.



**Solubility** (SAHL yoo BIL uh tee) is the ability of a substance to dissolve in another substance. Flavored drink mix dissolves in water.

**Ductility** (duhk TIL uh tee) is the ability of a substance to be pulled into a wire. Copper is often used to make wiring because it is ductile.



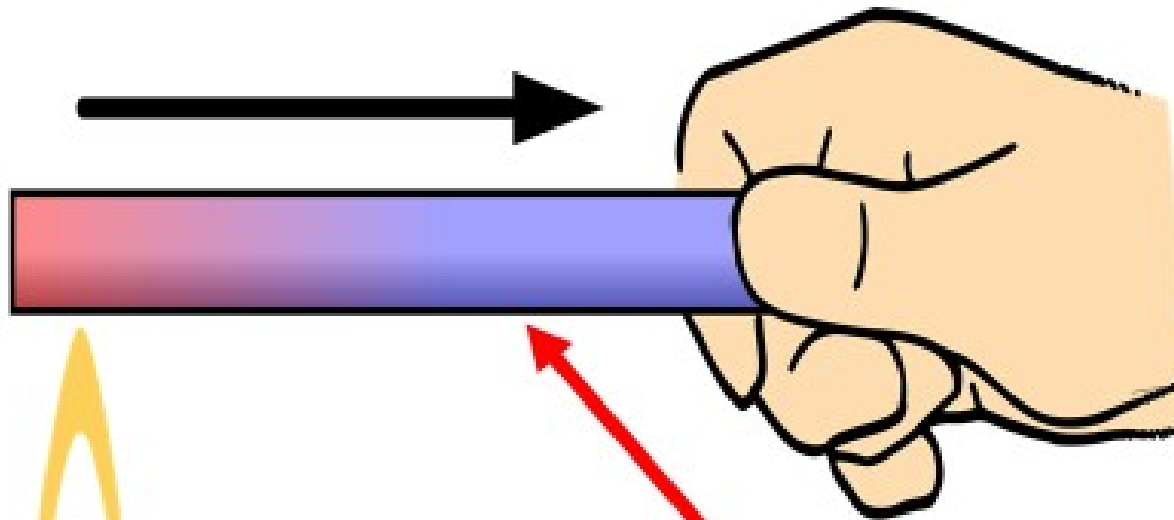
**Malleability** (MAL ee uh BIL uh tee) is the ability of a substance to be rolled or pounded into thin sheets. Aluminum can be rolled into sheets to make foil.



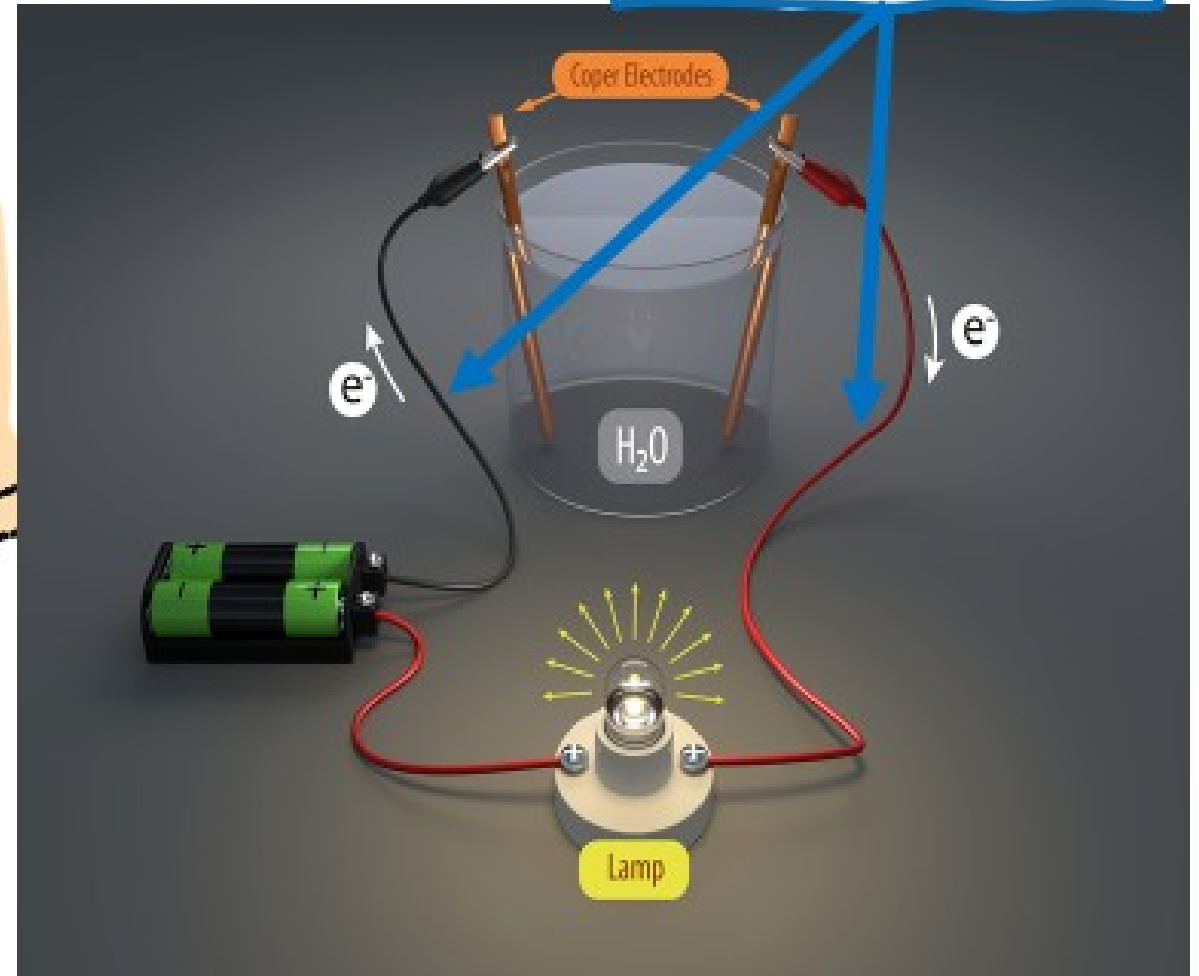
Pronounce and understand today's keywords

# Conduction

Electrical  
Conductor



Thermal  
Conductor



We gather all of the information that we need in order to learn.